EL NIÑO ACTION BRIEFING · 12 OCTOBER 2026

Chapalita, Guadalajara

The first hazard to arrive is not El Niño. A water cut caused by contamination began on 6 October. For Chapalita's older residents, the El Niño winter's cold matters more than the spring heat. Two things to do first: get every resident over 60 vaccinated in the winter campaign now running, and make the water you store safe.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
The Climate Impact Lab region containing Guadalajara municipality (name not resolved)	No figure published for Mexico	n/a (the global study runs June 2026 to February 2027)	Water cuts now · cold in winter · heat and smoke in spring

12/12 recommendations checked. 9 corrected, 1 unverified, 2 contraindicated. One circulating piece of advice was also found contraindicated and three were softened. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes no figure for Mexico or for any Latin American country except Brazil, so no death count is given here, and none has been scaled down from another country. **The funding target fell by 35% in verification**, from MXN 2.54 million to MXN 1.64 million. One placement was dropped, and a second moved above the target because its recipient and evidence could not be confirmed.

EXECUTIVE SUMMARY — AT A GLANCE



20.67°N 103.40°W, Mexico · Natural Earth

- **The risk.** Water cuts lead: Guadalajara's supply breaks when one link fails, as in 2021 (213 colonias rationed) and the 6 October 2026 contamination cut bringing six months of tandeos; winter cold then threatens older residents.
- **First action.** Vaccinate every resident 60 and over by 30 November 2026 (IMSS, ISSSTE, Secretaría de Salud Jalisco), with the neighbours' association phoning those living alone on cold nights through February 2027.
- **The money.** MXN 600,000 (about US$33,000) floor, MXN 1,640,000 (about US$90,000) target, MXN 2,540,000 (about US$140,000) ceiling; the top-ranked placement is DIF Guadalajara's isolated-elder register and daily contact, MXN 600,000.
- **Do not.** Do not say "drink more water" without naming a safe source: tap water is reported dark and foul-smelling in 475+ colonias, is being cut for contamination, and is untested.*

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* Suggested formulation: check with a health worker before sharing.

Scope. The request gave a street address in Guadalajara. A street address is too precise for this kind of briefing, and it is not where the public data exists, so this briefing covers the colonia of **Chapalita, Guadalajara (CP 44500)**. Chapalita is in Guadalajara municipality. The adjoining Chapalita Oriente and Chapalita Sur (CP 45040) are in Zapopan, so where an action runs through a municipality, both are named. Statistics below are for Jalisco state or the metro area unless they say otherwise. No colonia-level census profile for Chapalita could be retrieved.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. Before December, get every resident 60 or over vaccinated in the winter campaign now running, make home heating CO-safe, and phone older people living alone on cold-alert nights

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MANAGER | ADAPTATION INVENTORY: PRECEDENT, MEXICO, SEGURO POPULAR CUT COLD DEATHS

OWNER	BY WHEN	COST	PREVENTS
IMSS, ISSSTE and Secretaría de Salud Jalisco (vaccines); neighbours' association (reaching people)	30 November 2026; calls through February 2027	Vaccines already funded federally; calls are volunteer time	Winter cardiovascular and respiratory deaths, and CO poisoning

In Mexico, cold kills far more people than heat: about 14 times as many, and 98% of cold deaths are in people aged 35 or over (Wilson et al. 2024, *Science Advances*). Temperatures outside the comfortable range cause about 26,000 deaths a year in Mexico, 3.8% of all deaths. 92% of those deaths fall on days below 20 °C, which describes a normal Guadalajara winter. When Seguro Popular widened health coverage, it saved at least 1,600 lives a year from cold (Cohen and Dechezleprêtre 2022, *AEJ: Policy*). The IMSS 2026–27 winter campaign runs from 12 October 2026 to 3 April 2027 and offers flu, COVID, pneumococcal and RSV vaccines to people 60 and over. Carbon monoxide causes most of the acute winter deaths that are reported: 6 of the 7 recorded nationally in 2025–26. Those reported deaths are a tiny fraction of total cold deaths, most of which are heart and lung deaths. Protección Civil Jalisco issues frost and cold alerts but has no formal early-warning system.

* Suggested formulation: check with a health worker before sharing.

Corrections: the evidence that flu vaccination reduces deaths in older people is weaker than usually claimed. A 332,000-person trial (DANFLU-2, NEJM 2025) found high-dose vaccine did not significantly cut hospitalisation compared with standard dose. The national weather service (SMN) forecasts 56 cold fronts from September to May, 6 above normal, but it cautions that more fronts does not mean colder. During the 2015–16 El Niño, west-central Mexico had a milder winter. The 13 December 1997 snowfall in Guadalajara is real, but it is a single event. Cold mortality is about twice as high in lower-income households, so the effect in Chapalita is probably smaller. The action is cheap, already funded, and running, which is why it ranks first.

2. Follow SIAPA's tandeo calendar for Chapalita, and make the water you store safe: cover, screen, wash, and chlorinate the tinacos and cisterns you already have

PARTIALLY CONFIRMED | SKEPTIC · HISTORIAN · FRONTLINE · EMERGENCY MANAGER · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Households; neighbours' association relays SIAPA notices	Now; the cut runs about six months	Chlorine, lids and mesh: household-level	Gastrointestinal illness from stored water, and dengue mosquitoes breeding in open containers

On 6 October 2026 Governor Lemus announced that the Las Pintas and old Chapala system would stop supplying the city because of contamination from irregular discharges. That system supplies up to about 2.5 m³/s of the metro area's 10–11.5 m³/s. The first stage lasts about six months, with rotating cuts (tandeos) of one to two days of no water or low pressure. SIAPA and the municipalities were due to publish the tandeo programme on 12 October. On the day this briefing was written, no colonia list existed, so **whether Chapalita is affected could not be checked**. Residents have reported dark, foul-smelling tap water in at least 475 colonias, and the government has not released water-test results. Boil or treat stored water before drinking it. Request water trucks (pipas) only through SIAPA's official channels.

* Suggested formulation: check with a health worker before sharing.

Corrections: the free-pipa phone number from 2021 could not be verified for 2026, so it is not printed here. Jalisco's 2024 dengue year (20,907 cases and 66 deaths, the most in Mexico) is confirmed. The share of those in the metro area is not. "Cover and screen" follows the Secretaría de Salud's *lava, tapa, voltea y tira* campaign (wash, cover, turn over, throw out). Most Chapalita homes already have a tinaco or cistern, so this is about making existing storage safe, not buying more.

3. Build a list of residents 70 and over who live alone, are bedridden, or rarely go out, and pair each one with a neighbour who checks on them on cold nights, smoke days, water-cut days, and heat days

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST · FOOD | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
DIF Guadalajara with the Chapalita neighbours' association	List by 31 January 2027; contact through June 2027	About MXN 600,000 (estimate; see funding)	Deaths of isolated people that nobody notices in time

In a pooled analysis of six heat-wave studies (Bouchama et al. 2007), being confined to bed multiplied the odds of death by 6.44. Not leaving home daily multiplied them by 3.35. Regular social contact cut them to 0.40 of the baseline. The same contact can also carry groceries or a meal on a cut or alert day. No delivery service currently does this, so it would be new.

* Suggested formulation: check with a health worker before sharing.

Transferability: that evidence covers heat only, from 1,065 deaths in the US and Europe. Its use here for cold, smoke and water cuts is reasonable but not measured. No DIF Guadalajara or DIF Zapopan register of isolated or bedridden older adults, and no home-visit programme, was found. DIF Guadalajara's "Años con Vida" is centre-based and reaches about 500 people. So this list is new, not a top-up to an existing service. It fits under the metro area's 2020 WHO age-friendly certification. The figure of 54,000 people reached by the age-friendly programme, which one lens cited, could not be verified and is not used.

4. Reach the domestic workers, gardeners, drivers, and builders who commute into Chapalita through the households that employ them: a paid heat-day and smoke-day protocol, and IMSS enrolment

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Neighbours' association, through employer households	Before April 2027	Printing; the IMSS cost depends on wage and days worked (use the IMSS simulator)	Heat illness in the people most exposed here, most of whom have no social security

Jalisco has about 174,000 paid domestic workers, 83% of them women. More than 90% have no IMSS cover: about 3,056 were enrolled in 2023. Employers have been legally required to enrol domestic workers since 16 May 2023, with fines of MXN 2,346–41,058. The heat-day protocol is voluntary good practice: water, shade, a midday break, and an earlier finish, paid, on alert days. Give each worker a heat-illness card in Spanish. Many of these workers commute from Tonalá, Tlaquepaque and Tlajomulco. They are in nobody's neighbourhood WhatsApp group, so the route to them is the household that employs them.

* Suggested formulation: check with a health worker before sharing.

Corrections: the federal workplace heat standard (NOM-015-STPS-2001) covers workplaces with heat sources. It does not credibly apply to private homes, so it is not cited as a legal duty. Nationally, 75% of heat deaths are in people under 35 (Wilson et al. 2024). That group includes infants, the paper did not test occupation, and the figure is national, not local.

5. Prepare for smoke from La Primavera forest, March–May 2027: when an atmospheric alert or contingency covers Guadalajara or Zapopan, close windows, run one clean-air room, keep vulnerable people inside, and give outdoor workers fitted N95 masks

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Households; association relays SIMAJ and SEMADET alerts	Filters and masks stocked by 28 February 2027	Box fan plus MERV-13 filter, or a HEPA unit, per household	Heart and lung attacks during smoke episodes

The fire in Bosque La Primavera on 9 March 2025 burned between about 1,000 and over 2,100 ha, depending on the source. It triggered an *alerta atmosférica* for Zapopan, Tala and El Arenal, and closed schools there. The state's PRECA contingency plan escalates from pre-contingency to phase III on PM10 or ozone levels, and was last activated on 25 December 2025. A box fan with a MERV-13 filter delivers about 111 cubic feet per minute of clean air, comparable to a small commercial purifier (US EPA chamber study). Filters clog fast in smoke, so keep spares. A mask only protects if it fits.*

* Suggested formulation: check with a health worker before sharing.

Corrections: the 2025 fire alert covered Zapopan, not Guadalajara municipality, so the trigger here is any alert for Guadalajara or Zapopan monitoring stations. Filters remove particles, not ozone. The estimate of about 655 air-pollution deaths a year in the metro area comes from a secondary press report and is undated. The filter evidence comes from US chambers and homes, but the physics is the same everywhere.

6. Agree a numeric heat trigger for April–June 2027 that switches on the daily checks from action 3

PARTIALLY CONFIRMED · WEAK TRANSFER | PUBLIC HEALTH · HISTORIAN · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Protección Civil Jalisco and Guadalajara, with Secretaría de Salud Jalisco	31 March 2027	None beyond action 3	Heat deaths among frail people living alone; likely few here

The alert machinery already exists. The federal health ministry runs weekly heat surveillance during the hot season, SMN issues heat-wave advisories, and Protección Civil Jalisco publishes heat alerts for the metro area (for example, when 35 °C is forecast). What is missing is a written link from that alert to outreach for isolated older people. That link is the whole ask.

* Suggested formulation: check with a health worker before sharing.

Transferability is weak, and this is why the action ranks sixth. The usual evidence comes from Ahmedabad's Heat Action Plan, which avoided an estimated 1,190 deaths a year (95% CI 162–2,218; Hess et al. 2018). But Ahmedabad's lowest alert is 41 °C, above Guadalajara's all-time record of 40.5 °C (June 2023). The study's individual rate differences were not significant, and the effect disappeared when 2010 was excluded. Jalisco recorded 1–2 heat-stroke deaths in 2025, against 79 nationally. That surveillance counts heat stroke only and undercounts real heat mortality, but expected heat deaths in Chapalita are small.

7. Jalisco's finance secretariat sets up the committee and operating rules for the state disaster fund for municipalities (FOEADENM), with an explicit heat-emergency trigger

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: PRECEDENT, MEXICO, FONDEN SAVED LIVES (BENEFIT–COST ≥3.2)

OWNER	BY WHEN	COST	PREVENTS
Secretaría de la Hacienda Pública, Jalisco; Guadalajara applies as beneficiary	**Start now; unlikely to pay out before spring 2027**	**None: money already budgeted (MXN 71m in 2023)**	**Money that exists but cannot be released, as happened in 2023**

FOEADENM is a state fund created in 2018. In 2023 its MXN 71 million sat unused because it had no technical committee and no operating rules (Crónica, 23 September 2023). No record was found of rules being issued since. Mexico's indexed federal fund, FONDEN, reduced deaths by restoring access to health services after disasters, with a benefit–cost ratio of at least 3.2 (del Valle 2024, *AEJ: Policy*). FONDEN was abolished in 2021. This is the only pre-arranged public money in reach, and fixing it is a regulatory change, not new spending.

* Suggested formulation: check with a health worker before sharing.

Corrections: the fund belongs to the state, not to Guadalajara. Whether Jalisco's civil protection law lets heat be declared an emergency could not be confirmed. Payouts on record from the separate state fund, FOEDEN, are for floods only. Mexico's 2026–27 federal catastrophe cover (MXN 10.4 bn, about US$596m) covers earthquakes, volcanic eruptions, hurricanes and floods, and not heat or drought. UN CERF's anticipatory El Niño money went to Central America, not Mexico.

8. Cruz Roja Mexicana starts a simplified heat Early Action Protocol with the IFRC's anticipatory fund (DREF), aimed at the 2028 hot season

PARTIALLY CONFIRMED · NOT BEFORE THIS PEAK | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Cruz Roja Mexicana national headquarters	Apply in 2027 for the 2028 season	About CHF 200,000 (the size of comparable heat protocols)	The next heat season arriving with no money pre-arranged

Simplified heat protocols exist in Greece (activated June 2024, CHF 63,900), Myanmar (urban heat, CHF 200,000), Burkina Faso (CHF 220,000) and Albania (CHF 200,000). The Mexican Red Cross has no protocol for any hazard on record.

Corrections: the national society applies, not the Jalisco delegation. In the pilot, only 21 of 46 simplified protocols had been approved by June 2025. Approval before March 2027 is unlikely, so this is preparation for the next event, not this one.

9. Watch two food-price series and act if they jump: SNIIM's Guadalajara tortilla price and INEGI's biweekly consumer price index (INPC) for Guadalajara

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: PRECEDENT, MEXICO, PROGRESA OFFSET DROUGHT'S EFFECT ON DIET

OWNER	BY WHEN	COST	PREVENTS
DIF Guadalajara; Banco de Alimentos de Guadalajara	Monthly from now to mid-2027	None	A price shock going unnoticed until households are already cutting meals

* Suggested formulation: check with a health worker before sharing.

The El Niño food signal here is weak and late. El Niño winters bring more rain to northern Mexico, which tends to help irrigated maize in Sinaloa. Tortilla prices follow the US maize price and the peso more than local rain. SNIIM publishes tortilla prices by city on Mondays, Wednesdays and Fridays. INEGI publishes the INPC for Guadalajara every two weeks. Suggested trigger: tortilla above MXN 28/kg, or Guadalajara food inflation two points above the national rate.

Corrections: the latest Guadalajara tortilla price found was MXN 25.85/kg (November 2025), so the MXN 28 trigger rests on old data. Recheck the current price before using it. Chapalita households spend far less of their budget on food than the national 37.7% (ENIGH 2024), so the people this protects are mainly the workers who commute in.

10. Build a drought reserve into SIAPA's water tariff, released when Lake Chapala's storage falls below a set level

UNVERIFIED ECONOMIST ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
SIAPA board, municipal councils, Jalisco Congress	The 2028 revenue law (the 2027 cycle has closed)	Negative: collected while reservoirs are full	The 2028 dry season, if El Niño cuts the 2027 rains, arriving with no reserve

SIAPA's tariff board proposes rates, the councils review them, and the state Congress approves them in the revenue laws. No reserve mechanism was found, and Congress opposed a rate rise for 2026. Water storage is high now (Chapala about 70%, Calderón about 88% at the end of September 2026). This is therefore the cheapest moment to write the trigger, but nothing about it is in place.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Neighbours' association & households	DIF Guadalajara	Salud Jalisco, IMSS & ISSSTE	SIAPA	Protección Civil (Jalisco & Guadalajara)	SEMADET	Jalisco Hacienda	Cruz Roja Mexicana	Banco de Alimentos	Jalisco Congress
1. Vaccines, CO-safe heating, cold-night calls	**R**	**C**	**A/R**		**R**					
2. Tandeo calendar; safe stored water	**R**		**C**	**A/R**	**I**					
3. Register of isolated residents 70+	**R**	**A/R**	**C**		**I**					
4. Commuting workers: heat days, IMSS	**A/R**		**C**							
5. Smoke days: clean room, N95	**R**		**C**		**C**	**A/R**				
6. Numeric heat trigger	**I**	**R**	**R**		**A/R**					
7. FOEADENM rules, heat trigger					**C**		**A/R**			**I**
8. Heat early-action protocol (2028)					**C**			**A/R**		
9. Food-price watch		**A/R**							**R**	
10. SIAPA drought reserve				**R**						**A/R**

What the chart shows: **the neighbours' association does the work in five rows and is accountable for one**. It is the only body here that reaches both the isolated residents and the households that employ the commuting workers. **DIF Guadalajara owns the register**, which has to be built new. **The three state-level rows (7, 8 and 10) need decisions, not money**, and none of them will move cash before this peak.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

MXN 600,000 (about US$33,000)

TARGET

MXN 1,640,000 (about US$90,000)

CEILING

MXN 2,540,000 (about US$140,000)

LARGEST UNCERTAINTY

Whether Chapalita is on the tandeo list, and how many people 70+ live alone there

Arithmetic, at about MXN 18.2 per US$ (Banxico FIX, 8 October 2026). Floor: the older-adult register, 2 coordinators × MXN 25,000 × 6 months (300,000), plus 40 volunteer stipends × MXN 1,500 × 5 months (300,000), giving MXN 600,000. Target: the floor plus drinking water, 500 older households × 2 jugs of 20 litres a week × 26 weeks × MXN 40, which is MXN 1,040,000, for a total of MXN 1,640,000. Ceiling: the target plus

* Suggested formulation: check with a health worker before sharing.

cash for 300 commuting workers × MXN 3,000 (about 9.5 days at the 2026 minimum wage of MXN 315.04), for a total of MXN 2,540,000. **The target fell 35% in verification**, from MXN 2,540,000: the ambulance top-up was dropped, and the worker cash was moved to the ceiling. The water placement only applies if Chapalita turns out to be rationed. Without it, the target is the floor.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
DIF Guadalajara, with the Chapalita neighbours' association	Action 3: the register and daily contact	MXN 600,000	Matching grant	Philadelphia's heat-warning system cost about US$210,000 over 1995–98 and was credited with 117 lives among people 65+ (Teisberg, Ebi et al. 2004, *BAMS*). That was a citywide system, not a buddy scheme alone, and the authors call 117 probably overstated. No DIF home-visit operation exists yet.	PARTIALLY CONFIRMED
Banco de Alimentos de Guadalajara	Action 2: drinking water for older households, only if Chapalita is rationed	MXN 1,040,000	Pre-positioned stock	Unit arithmetic as shown. The MXN 40 jug price is unverified. The food bank serves more than 135,000 people in 361 communities, but it delivers food, not water.	PARTIALLY CONFIRMED, CONDITIONAL
Cáritas de Guadalajara, through parishes (ceiling only)	Action 4: cash for commuting workers' lost heat-day wages	MXN 900,000	Unconditional cash transfer	No current Cáritas cash-transfer activity was found. The Mexican income-support evidence (Sarmiento et al. 2024) concerns a lasting wage policy, not one-off heat-day payments.	UNVERIFIED
~~Cruz Roja Mexicana, Jalisco~~	~~Extra ambulance shifts~~	~~MXN 400,000~~	Top-up	No cost basis and no capacity figures were found.	DROPPED

The funder's one thing: expected heat deaths in Chapalita itself are probably close to zero. The money that saves lives for "Chapalita's people" goes to the isolated older residents inside it and to the workers who come in each day. Spending to make the colonia itself ready is poor value. Money above the ceiling does more in the rationed eastern colonias, such as Oblatos and Miravalle.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here, or would spend the window on something that cannot work before the peak.

“Drink more water” with no safe source named

CONTRAINDICATED

Standard heat advice says to drink more water. Here, residents report dark, foul-smelling tap water in at least 475 colonias, the supply is being cut because of contamination, and the government has not released test results. Advice to drink more, with no source named, sends people to the tap.

Instead: every hydration message names a safe source: a sealed garrafón (water jug), boiled water, or treated water.*

Store water in open or uncovered containers

CONTRAINDICATED AS WORDED; STORAGE ITSELF IS NECESSARY

During a six-month tandeo, households will store water. Uncovered storage gives Aedes mosquitoes somewhere to breed, and Jalisco led Mexico in dengue in 2024 (20,907 cases, 66 deaths). Storage itself is not the danger: the Oaxaca study usually cited has no data comparing covered and uncovered containers. One lens’s advice to “store water now” was therefore softened, not dropped.

Instead: cover, screen, and wash what you already have, and add container checks to dengue brigades before the rains return.

Count on an early pension payment or heat insurance to pay out before this spring

CONTRAINDICATED: WASTES THE WINDOW

Two financing ideas the panel raised cannot move money before April 2027. Every early payment of the older-adult pension found (2019, 2024) was tied to an election blackout, and no route exists to pay it early because of heat. No urban heat-income insurance exists in Mexico. A trigger of 36 °C for three days was very likely crossed in 2023 and 2024, which would make the premium unaffordable. India’s

* Suggested formulation: check with a health worker before sharing.

working scheme pays about US$2–20 per member.

Instead: spend the effort on actions 3, 4 and 7. If insurance is wanted, scope it as a study for 2028.

Softened, not contraindicated. “Don’t use fans above 35 °C” is stricter than the evidence. Morris et al. 2021 (*Lancet Planetary Health*) put the point where fans stop helping at about 38 °C for people over 65, and about 37 °C for over-65s taking anticholinergic medicines. Guadalajara rarely goes above about 36 °C, so fans help here, with that caveat for older people on those medicines. “Save water for the El Niño drought” is mistimed this winter, because the current cut is about quality, not shortage. If El Niño weakens the 2027 rains, drought messaging belongs to spring 2027.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1955	Lake Chapala at its record low (1 July: 954 Mm³)	Not recorded	The lake refilled within wet seasons, as it has every time since
Dec 1997	Snow across the metro area during the 1997–98 El Niño, up to about 3.5 cm in south Zapopan, the first since about 1881	No casualty figure found	Nothing found
1998	One of La Primavera’s three largest fires	Area and smoke not verified	Nothing found
2002	Chapala’s second-lowest level (6 June: 1,138 Mm³)	Not recorded	Aqueduct and dam projects promised; El Zapotillo stalled by court injunctions for two decades
2021	Calderón dam near empty; SIAPA rationed 213 colonias	No deaths recorded; many homes without water for up to two weeks	Free water trucks; UdeG researchers blamed late action, not the drought itself

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
June 2023	Metro-area heat record, 40.5 °C	Jalisco heat deaths for 2023 not found	Nothing found
2024	Chapala down to about 46%; record dengue year	Dengue: 20,907 cases, 66 deaths in Jalisco	El Salto–La Red–Calderón aqueduct running since 2024 (about 1 m³/s)
Mar 2025	La Primavera fire, about 1,000–2,100 ha	No deaths reported; schools closed in parts of Zapopan	Atmospheric alert issued
Oct 2026	Las Pintas / old Chapala supply cut for contamination; about six months of tandeos	Not yet known	Protests; a new MXN 18 bn Chapala–Guadalajara aqueduct proposed

The recurrence nobody has noticed: Guadalajara answers each water crisis with a new pipeline that arrives after the rains have already ended it. The lake has refilled within one or two wet seasons every time since 1955. Supply to colonias is actually cut when one link in the delivery chain fails: a dam, a pumping station's transformers, or now a contaminated canal. The window to watch this cycle is May–June 2027, when peak heat, fire smoke and peak water demand land on that chain together.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Store water now** (Historian) versus **storage breeds dengue** (Skeptic). Resolved here as: store, but cover and screen. Settled by data comparing dengue breeding in covered and uncovered containers in Guadalajara, which was not found.
- **El Niño is the frame** (Emergency manager, Economist: cold this winter, drought in 2028) versus **El Niño is not the main threat** (Skeptic: the contamination cut and the normal April–June heat matter more). Both are partly right. This briefing ranks the live water cut and the cold first, whatever their cause.
- **Plan for heat** (Public health, Historian: a heat action plan) versus **plan for cold** (Public health's own finding: in Mexico, cold kills older people and heat kills younger ones). Settled locally by Jalisco cause-of-death data by season and age, which no lens retrieved.
- **Does El Niño make this winter colder?** SMN forecasts more cold fronts and there was snow in 1997, but the 2015–16 El Niño gave west-central Mexico a milder winter. The evidence is mixed; action 1 is worth doing either way.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

No lens assessed **what is in the tap water**. The supply is being cut because of contamination, residents report dark, foul-smelling water, and no test results have been published. If the contaminant is something boiling does not remove (metals, industrial chemicals), the household advice in action 2 is not enough, and the drinking-water placement becomes the top priority rather than a conditional one. Two smaller gaps: no lens could find Chapalita's age structure or how many homes have air conditioning, and none looked at **power cuts** during heat, which would knock out fans and electric pumps for cisterns at the same time.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The request gave a street address in Guadalajara. It was widened to its colonia: Chapalita, Guadalajara (CP 44500), located with OpenStreetMap. Climate Impact Lab impact region, as the report records it: the region containing Guadalajara municipality (name not resolved). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. The study page and report PDF name no figure for Mexico, so none is given. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.

* Suggested formulation: check with a health worker before sharing.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in English and Spanish, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. This demoted the heat action plan: several lenses backed it, but its evidence comes from Ahmedabad, which is much hotter. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted by reading the dataset embedded in the inventory page. It holds eight Mexico entries: four on conditional cash transfers (PROGRESA/Oportunidades), Seguro Popular's effect on cold mortality, a minimum-wage and VAT reform, public health insurance for schoolchildren, and FONDEN. The precedents are tagged on actions 1, 7 and 9. It has no Mexico entries on wildfire smoke, water rationing, heat warnings, or outreach to isolated people.
- **Health wording.** Passages where a wording error could hurt someone (safe drinking water and wildfire smoke) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Run-specific limits:
  - The impact region name was not resolved.

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- SIAPA's tandeo colonia list had not been published on the run date.
- The state statistics institute's (IIEG) colonia profile for Chapalita could not be retrieved.
- Jalisco heat deaths for 2023–24 were not found.
- Several figures rest on Mexican press reports rather than primary data. They are marked where that matters.
- Whether Guadalajara is on CFE electricity tariff 1 or 1C was not resolved.

References · Wilson et al. 2024, Science Advances (verified) · Cohen & Dechezleprêtre 2022, AEJ: Policy (verified) · IMSS winter vaccination 2026–27 (dates from search text) · SMN cold fronts, Crónica (verified) · UdeG on the 2015–16 winter · Winter deaths 2025–26, Excélsior · Las Pintas cut, La Jornada, Infobae, Informador (verified) · Dark tap water, El Imparcial · SIAPA discount, El Financiero · Dengue 2024, Azteca Jalisco (verified) · Storage and Aedes, PMC7068688 · Reservoir levels, Azteca Jalisco · 2021 rationing, La Jornada · Bouchama et al. 2007, Arch Intern Med (verified) · Teisberg, Ebi et al. 2004, BAMS (verified) · Hess et al. 2018, J Environ Public Health (verified) · PAHO age-friendly certification · DIF Guadalajara Años con Vida · Heat deaths 2025, UnoTV · Heat alert example, Azteca Jalisco · Domestic workers, Jalisco Igualdad, IMSS (verified) · NOM-015-STPS-2001 · La Primavera 2025, N+ · SIMAJ · PRECA Dec 2025, La Razón · EPA DIY air cleaner, PMC9828579 · FOEADENM, Crónica · Federal cat cover, Reinsurance News (verified) · CERF, PreventionWeb · Heat sEAPs, Greece, Myanmar · Pension, Fox Sports MX · SIAPA tariffs, Crónica · SNIIM tortilla prices · IIEG Guadalajara inflation · Minimum wage 2026, UnoTV · Exchange rate, ViveUSA · Morris et al. 2021, University of Sydney summary · Food: SNIIM and press via the food lens; Sinaloa maize, eggs and cold-front items are as reported, not independently verified
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 12 October 2026 for Chapalita, Guadalajara, Jalisco. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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